

Lecture 19:

Monday, Sep 12, 2022

Theorem 1: If  $\phi: G_1 \rightarrow G_2$  is an isomorphism, then  $\phi^{-1}: G_2 \rightarrow G_1$  is also an isomorphism.

Proof: Let  $x, y \in G_2$ . Then, there exist unique  $a$  and  $b$  in  $G_1$  such that  $\phi(a) = x$  and  $\phi(b) = y$ .

Then,  $a = \phi^{-1}(x)$  and  $b = \phi^{-1}(y)$

Now,  $xy = \phi(a)\phi(b) = \phi(ab) \Rightarrow ab = \phi^{-1}(xy)$

$\Rightarrow \phi^{-1}(xy) = ab = \phi^{-1}(x)\phi^{-1}(y) \quad \forall x, y \in G_2.$

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Theorem 2: If  $\phi: G_1 \rightarrow G_2$  and  $\psi: G_2 \rightarrow G_3$  are group isomorphisms, then  $\psi \circ \phi: G_1 \rightarrow G_3$  is also a group isomorphism.

Proof: For  $x, y \in G_1$ , we have

$$\begin{aligned}(\psi \circ \phi)(xy) &= \psi(\phi(xy)) = \psi(\phi(x) \cdot \phi(y)) \\&= \psi(\phi(x)) \psi(\phi(y)) \\&= (\psi \circ \phi)(x) (\psi \circ \phi)(y)\end{aligned}$$

$\therefore \psi \circ \phi$  is a group homomorphism. Since  $\phi$  and  $\psi$  are bijective,  $\psi \circ \phi$  is a bijective. Hence,  $\psi \circ \phi$  is an isomorphism.

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Ex1: If  $G$  is a finite and  $f: G \rightarrow \mathbb{Z}$  is a homomorphism, then  $f$  is the trivial homomorphism, that is,  $f(x) = 0 \forall x \in G$ .

Ex2: Find all the group homomorphisms  $f: \mathbb{Z}_n \rightarrow \mathbb{Z}_m$

Solution:  $f(k) = k f(1)$ , so  $f$  is determined by the value of  $f(1)$ . Hence,  $f(k) = a \cdot k$  for some fixed  $a \in \mathbb{Z}_m$ .

For  $a \in \mathbb{Z}_m$ , let  $f_a: \mathbb{Z}_n \rightarrow \mathbb{Z}_m$  be defined by

$$f_a(x) = ax \quad \forall x \in \mathbb{Z}_n$$

If  $f_a: \mathbb{Z}_n \rightarrow \mathbb{Z}_m$  is a group homomorphism,  
then

$$0 \equiv f_a(0) \equiv f_a(n) \equiv n \cdot f_a(1) \equiv n \cdot a \pmod{m}.$$

By Theorem 1 ( $b=0$  case) of Lecture 8, we have

the congruence  $nx \equiv 0 \pmod{m}$  has  $d = \gcd(n, m)$   
solutions, namely,  $x = \frac{m}{d}k$ ,  $k = 0, 1, 2, \dots, d-1$ .

For each solution  $x \equiv a$ , we have a homomorphism  
 $f_a: \mathbb{Z}_n \rightarrow \mathbb{Z}_m$ .

Hence, the set of homeomorphisms  $f: \mathbb{Z}_n \rightarrow \mathbb{Z}_n$  is

$$\{f_a \mid a = \frac{m}{d} k, \quad k = 0, 1, \dots, d-1\}$$

Here  $f_a: \mathbb{Z}_n \rightarrow \mathbb{Z}_n$  is the map defined by

$$f_a(x) = ax \quad \forall x \in \mathbb{Z}_n.$$

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## § Automorphisms:

$\text{Aut}(G) = \{ \phi : G \rightarrow G \mid \phi \text{ is an automorphism} \}$  is a group under composition of functions.

Ex 1:  $\text{Aut}(\mathbb{Z}) = \{ I, \phi \} \cong \mathbb{Z}_2$ , here  $\phi(x) = -x \ \forall x \in \mathbb{Z}$ .  
In fact, if  $G$  is an infinite cyclic group, then  $\text{Aut}(G) \cong \mathbb{Z}_2$ .

Ex 2: If  $G$  is a finite cyclic group of order  $n$ , then

$$G \cong (\mathbb{U}(n), \cdot).$$

$$\text{Ex 3: } \text{Aut}(\mathbb{Q}) \cong (\mathbb{Q}^*, \cdot)$$

Let  $x = \frac{p}{q}$ ,  $x \neq 0$ ,  $q > 0$ .

Then, if  $f: \mathbb{Q} \rightarrow \mathbb{Q}$  is a homomorphism, we have

$$q \cdot f(x) = f(q \cdot x) = f(p) = p \cdot f(1)$$

$$\Rightarrow f(x) = \frac{p}{q} \cdot f(1) = x \cdot f(1).$$

If  $x = 0$ , then  $f(x) = 0 = x \cdot f(1)$ .

$$\therefore f(x) = x \cdot f(1) \quad \forall x \in \mathbb{Q}.$$

$$\text{Thus, } \text{Aut}(\mathbb{Q}) = \{f_a \mid a \in \mathbb{Q}, a \neq 0\} \cong (\mathbb{Q}^*, \cdot). \quad \#$$

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